Sadman Sadeed Omee

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Summary

- **Ph.D. Candidate on machine learning (ML) research with 4+ years of experience** in graph neural networks (GNNs), generative AI (e.g., large language models (LLMs) and diffusion models) and foundation models.
- **12+ publications (550+ citations)** in top journals, such as NPJ Computational Materials (Nature Portfolio), Advanced Science, and Patterns (Cell Press).
- **Industry experience** building production-oriented ML systems at KLA and Lawrence Livermore National Laboratory.
- **Research focus:** GNNs, AI for Science, Generative AI, Foundation Models, and Geometric Deep Learning

Experience

University of South Carolina **January 2022 – Present**
Graduate Research Assistant | *Machine Learning and Evolution Laboratory* Columbia, SC, United States

- **Designed and applied advanced deep learning models**, including graph neural networks (GNNs), large language models (LLMs), and diffusion models, to solve problems from the materials science domain such as crystal structure prediction (CSP), materials property prediction, and generative models for materials.
- **Authored multiple peer-reviewed research publications** in high-impact journals, demonstrating strong research productivity and impact in AI for science.
- **Currently developing a conditional E(3)-equivariant diffusion model** for 3D crystal polyhedra generation from chemical compositions, targeting fundamental scalability and accuracy bottlenecks in CSP.

KLA **May 2025 – August 2025**
Summer Intern Ann Arbor, MI, United States

- **Developed a continual transfer learning framework** for predicting semiconductor wafer critical dimensions from spectral data.
- **Improved out-of-distribution (OOD) robustness of ML-based metrology models**, enabling reliable performance under real-world manufacturing distribution shifts
- **Collaborated with cross-functional engineering teams** in a production-oriented research environment.

Lawrence Livermore National Laboratory **May 2024 – August 2024**
Summer Intern Livermore, CA, United States

- **Co-developed a multimodal molecular foundation model** integrating molecular structure, XANES spectra, SMILES representations, and HOMO–LUMO gap data using contrastive learning–based latent space alignment.
- **Implemented modality-specific encoders**, including GNNs for molecular structures and transformers for spectral and string-based modalities, to enable unified cross-modal molecular representations through self-supervised approaches.
- **Participated in weekly research seminars** with LLNL scientists focused on advanced machine learning models for materials and molecular science, high-performance computing (HPC), and related computational science fields.

Education

University of South Carolina **August 2021 – Present**
Ph.D. in Computer Science | GPA: 3.97/4.00 | **Expected Graduation:** August 2026 Columbia, SC, United States

University of South Carolina **August 2021 – December 2024**
M.S. in Computer Science (en route to Ph.D.) | GPA: 3.97/4.00 Columbia, SC, United States

Bangladesh University of Engineering and Technology **February 2015 – April 2019**
B.S. in Computer Science and Engineering Dhaka, Bangladesh

Skills

- **Deep Learning:** PyTorch, TensorFlow, Scikit-Learn, PyTorch Lightning
- **Graph & Geometric ML:** GNN, Equivariant Neural Networks, PyTorch Geometric
- **Generative AI & Foundation Models:** Diffusion Models, Transformers, LLMs, RAG
- **AI for Science:** Pymatgen, ASE, Matminer, RDKit, Atom3D
- **CV & NLP:** CNN, OpenCV, Hugging Face, NLTK, spaCy
- **Programming Languages:** Python, C, C++
- **Systems & Tooling:** Docker, Git, Bash

Selected Publications

1. **Omee, S. S.**, Wei, L., Dey, S., & Hu, J. (2025). Polymorphism crystal structure prediction with adaptive space group diversity control. ***Advanced Science***.
   - Devised **ParetoCSP2**, a deep neural network interatomic potential-based algorithm for crystal polymorph prediction that incorporates an adaptive space group diversity control, achieving up to 8.6× performance improvement and faster convergence.
2. Fu, N., Wei, L., Song, Y., Li, Q., Xin, R., **Omee, S. S.**, Dong. R., & Hu, J. (2023). Material transformers: Deep learning language models for generative materials design. ***Machine Learning: Science and Technology***.
   - Trained seven transformer **large language models (LLMs)** for generative materials composition design, achieving high chemical validity, novelty, and tunability, demonstrating their potential for discovering new materials.
3. **Omee, S. S.**, Louis, S. Y., Fu, N., Wei, L., Dey, S., Dong, R., Li, Q., & Hu, J. (2022). Scalable deeper graph neural networks for high-performance materials property prediction. ***Patterns (Cell Press)***.
   - Developed **DeeperGATGNN**, a global-attention-based GNN for materials property prediction that can leverage long-range atomic information, achieving improved performance and scalability (> 50 graph convolution layers) over existing models.
4. **Omee, S. S.**, Fu, N., Dong, R., Hu, M., & Hu, J. (2024). Structure-based out-of-distribution (OOD) materials property prediction: a benchmark study. ***NPJ Computational Materials (Nature Portfolio)***.
   - Developed a **comprehensive benchmark** for out-of-distribution (OOD) materials property prediction, revealing a significant performance gap for current GNNs in predicting properties of novel exceptional materials.
5. **Omee, S. S.**, Wei, L., Hu, M., & Hu, J. (2024). Crystal structure prediction using neural network potential and age-fitness pareto genetic algorithm. ***Journal of Materials Informatics***.
   - Developed **ParetoCSP**, a novel algorithm for crystal structure prediction that integrates a genotypic age-fitness criterion enhanced multi-objective genetic algorithm with a deep neural network inter-atomic potential.

Research Metrics

• **Citations:** 559 • **h-index:** 11 • **i10-index:** 12 (Last updated: March 13, 2026)

Selected Academic Projects

- **Foundation Model for In-Context Learning:** Developed a structure-aware in-context learning **foundation model** framework for materials property prediction, with ongoing publication work. (PyTorch, PyTorch Geometric, TabPFN)
- **Facet:** Developed an efficient **E(3)-equivariant GNN** that accelerates the training of foundation models for interatomic potentials by over an order of magnitude while maintaining comparable performance. (PyTorch, PyTorch Geometric)
- **FedMat:** Developed a full pipeline for **federated learning** for the out-of-distribution materials property prediction problem using graph neural networks. (FedChem, GNNs, PyTorch, PyTorch Geometric)
- **DeepXRD:** Developed a **convolutional neural network (CNN)** for the challenging task of XRD spectra prediction of crystals given their chemical compositions. (CNN, PyTorch, Pymatgen)
- **Election Information Quality Analysis:** Analyzed the quality of official information available for elections in 2024 across 12 US states based on multiple criteria using **NLP techniques**. (NLTK, spaCy, Transformers, LLMs)
- **Human Activity Recognition:** Developed a **spiking neural network (SNN)** for the problem of automatically recognizing and understanding the activities performed by a person from sensor data. (Tensorflow, SNN, Scikit-Learn)

Services and Memberships

- **Reviewing Services:** More than 35 peer reviewing services for several prestigious journals including *Advanced Science*, *PLOS ONE*, *npj Computational Materials*, *The Innovation*, *Pattern Recognition*, *Neural Networks*, and *Heliyon*.
- **Mentoring Services:** Mentoring undergraduate students in AI/ML research through Bangladeshi Student Research Initiative (**BSRI**, website: `https://sites.google.com/view/bsri-bd/team`).
- **Notable Memberships:** 1. Institute of Electrical and Electronics Engineers (**IEEE**): 2025 – Present & 2. American Association for the Advancement of Science (**AAAS**): 2025 – Present.